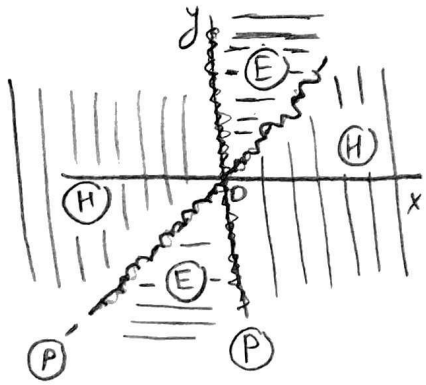
• For Elliptic, $\Delta < 0$: $\Rightarrow$

$\Rightarrow \Delta = 4x^2 - 4yx < 0 \Rightarrow 4x(x-y) < 0$

(E) $\boxed{x < 0 \ \& \ y < x}$ or (E) $\boxed{x > 0 \ \& \ y > x}$




Prob 5. (2 & T, Ch. 5, 5.1)

Let $P$ be the nonlinear operator in $\mathbb{R}^2$ defined by

(1) $$P(u) = \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}\right)^2 - u^2.$$

Show that each of the ↓ f-ns

(2) $$u_1(x,y) = e^x, \quad u_2(x,y) = e^{-y}$$

are sol-ns of the homogenous eq-n

(3) $$P(u) = 0$$

while their sum

(4) $$u(x,y) = e^x + e^{-y}$$ is not a sol-n.

Does this violate the principle of superpos-n?

Sol-n: [Preamble].

$$P(u_1) = \left(\frac{\partial}{\partial x}[e^x] + \frac{\partial}{\partial y}[e^x]\right)^2 - [e^x]^2 = 0 \quad \text{by (1) \& (3).}$$

$$\Rightarrow (e^x + 0)^2 - e^{2x} = e^{2x} - e^{2x} \equiv 0 \checkmark, \text{ so } u_1 = e^x$$

is a sol-n for (2).

Similarly,

$$P(u_2) = \left(\frac{\partial}{\partial x}[e^{-y}] + \frac{\partial}{\partial y}[e^{-y}]\right)^2 - [e^{-y}]^2 = 0 \quad \text{by (2) \& (1).}$$

$$\Rightarrow \left(0 - e^{-y}\right)^2 - e^{-2y} = e^{-2y} - e^{-2y} = 0 \checkmark, \quad \text{hence}$$

$u_2(x,y) = e^{-y}$ is a sol-n for (3), too.

Whereas, their linear combination (for superposition principle)

$u(x,y) = e^{x} + e^{-y}$ is NOT a sol-n for (3) because

$$P(u) = \left(\frac{\partial}{\partial x}[e^{x} + e^{-y}] + \frac{\partial}{\partial y}[e^{x} + e^{-y}]\right)^2 - [e^{x} + e^{-y}]^2 \overset{?}{=} 0$$

$$\Rightarrow (e^{x} - e^{-y})^2 - [e^{x} + e^{-y}]^2 = -4e^{x}e^{-y} \neq 0. \quad \text{So}$$

to answer if this violates Prin. of superp-n, recall Thm 5.5.9.

Let $P(\underline{x}, D) = \sum_{|\alpha| \le m} a^{\alpha} D^{\alpha}$ be a <u>linear operator</u>.

(for the <u>linear</u> PDE $\Rightarrow$ $a^{\alpha}$ coef-ts can NOT be a f-n of dependent variable)

Then, $P(c_1 u_1 + c_2 u_2) = c_1 P(u_1) + c_2 P(u_2)$.

Considering the given operator in (1) is NON linear, this operator can not be applied for Thm 5.5.9, hence the superpos-n principle is <u>not violated</u>. ∎

<u>Prob. 6.</u> (Z&T, Ch.5, 9.2)

Let $P$ be a <u>linear</u> PD operator.

(a) Show that if $u_1$ & $u_2$ are sol-ns to the nonhomogeneous eq-ns

(1) $$P(u_1) = f_1, \quad P(u_2) = f_2,$$

then $u = c_1 u_1 + c_2 u_2$ is a sol-n of the eq-n

$$P(u) = c_1 f_1 + c_2 f_2.$$